CMPG 325 — COMPUTER NETWORKS

MILESTONE 2: NETWORK IMPLEMENTATION, CONFIGURATION AND TESTING

Project ID: CMPG325-2026-063
Client: Phenyo Cultural Tours
Location: Potchefstroom, South Africa
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1. INTRODUCTION

This document presents the implementation, configuration and testing of the computer network designed for Phenyo Cultural Tours as part of the CMPG 325 individual semester project.

The organisation requires a reliable network to support its Booking and Reservations, Tour Operations, Finance, Marketing and server infrastructure, together with wireless connectivity for staff and guests.

The network was implemented and simulated using Cisco Packet Tracer. The implementation includes a multilayer core switch, four departmental access switches, separate VLANs, inter-VLAN routing, centralised DHCP and DNS services, a web server, wireless networking and switch security configurations.

The project also addresses the client's security requirements through secure device administration and switch port security. The IP addressing plan was designed using Variable Length Subnet Masking (VLSM) to accommodate the required users and support future expansion.

The purpose of this milestone is to document the implemented network, describe the configurations applied, and present the available evidence used to verify network connectivity, services and security.

2. NETWORK IMPLEMENTATION

2.1 Physical topology

The network uses a centralised topology in which CORE-L3 acts as the main switching and routing device. The four departmental access switches connect end devices to the network, while the servers provide centralised network services.

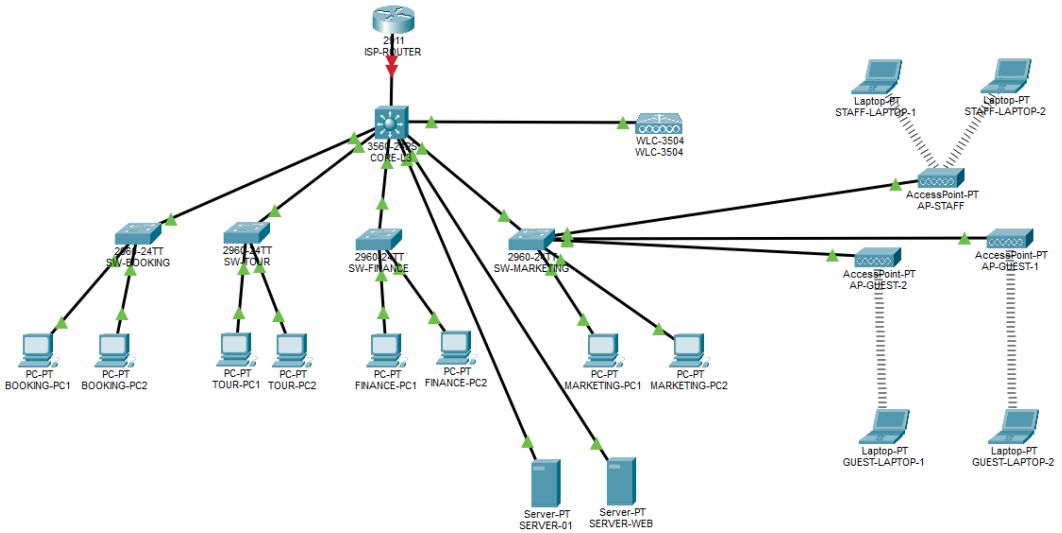
The main devices include:

- CORE-L3 — Cisco Catalyst 3560 multilayer switch.
- SW-BOOKING — Booking departmental access switch.

- SW-TOUR-OPS — Tour Operations departmental access switch.
- SW-FINANCE — Finance departmental access switch.
- SW-MARKETING — Marketing departmental access switch.
- SERVER-01 — Centralised DHCP and DNS server.
- SERVER-WEB — Web server hosting the organisation's website.
- WLC-3504 — Wireless LAN controller.
- Wireless access points and wireless client devices.

The access switches provide connectivity to departmental computers and other end devices. Trunk links carry traffic for the required VLANs between the access switches and the core switch.

Figure 1: Overall network topology



The topology separates departmental traffic logically while allowing the core switch to route traffic between configured VLANs where required.

2.2 Logical network design

The network is divided into VLANs to organise devices according to their functions.

VLAN ID	VLAN name	Subnet	Default gateway
10	Booking	172.30.38.0/26	172.30.38.1
20	Tour Operations	172.30.38.64/26	172.30.38.65
30	Finance	172.30.38.128/27	172.30.38.129
40	Marketing	172.30.38.160/27	172.30.38.161
80	Guest Wi-Fi	172.30.39.0/25	172.30.39.1
99	Servers	172.30.38.192/27	172.30.38.193
1	Management	172.30.39.128/25	172.30.39.129

The VLAN configuration allows the organisation to manage its departments as separate logical networks. The Finance department can be managed separately from other departments, while guest wireless devices use their own VLAN.

Figure 2: VLAN configuration evidence

```
WARNING: Authorised access only!

User Access Verification

Password:

CORE-L3>enable
Password:
CORE-L3#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/5, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   BOOKING                          active
20   TOUR-OPS                         active
30   FINANCE                          active
40   MARKETING                        active
80   GUEST-WIFI                       active
99   SERVERS                          active    Fa0/7, Fa0/8
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
CORE-L3#
```

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2.3 IP addressing and VLSM

The assigned private IPv4 address block is 172.30.38.0/23, which contains 512 total addresses and 510 usable host addresses.

VLSM was used to allocate different subnet sizes according to the needs of each network segment. The Booking and Tour Operations departments use /26 subnets, while Finance, Marketing and Servers use /27 subnets. Guest Wi-Fi and Management each use /25 subnets.

A separate /27 subnet was reserved for future expansion.

This addressing design supports efficient address allocation and provides room for the planned increase in users without renumbering existing departmental networks.**Figure 3:**

Routing table verification

```
CORE-L3#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     172.30.0.0/16 is variably subnetted, 7 subnets, 3 masks
C       172.30.38.0/26 is directly connected, Vlan10
C       172.30.38.64/26 is directly connected, Vlan20
C       172.30.38.128/27 is directly connected, Vlan30
C       172.30.38.160/27 is directly connected, Vlan40
C       172.30.38.192/27 is directly connected, Vlan99
C       172.30.39.0/25 is directly connected, Vlan80
C       172.30.39.128/25 is directly connected, Vlan1

CORE-L3#
```

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2.4 Inter-VLAN routing

Inter-VLAN routing was configured on CORE-L3 using switched virtual interfaces (SVIs). Each configured VLAN has a gateway address used by devices to communicate outside their local subnet.

IP routing enables the multilayer switch to forward traffic between the configured VLANs. DHCP relay was configured on VLANs 10, 20, 30, 40 and 80 so that client requests can reach the central DHCP server at 172.30.38.195.

The routing table was examined to verify the presence of the configured network routes.

3. NETWORK SERVICES

3.1 DHCP configuration

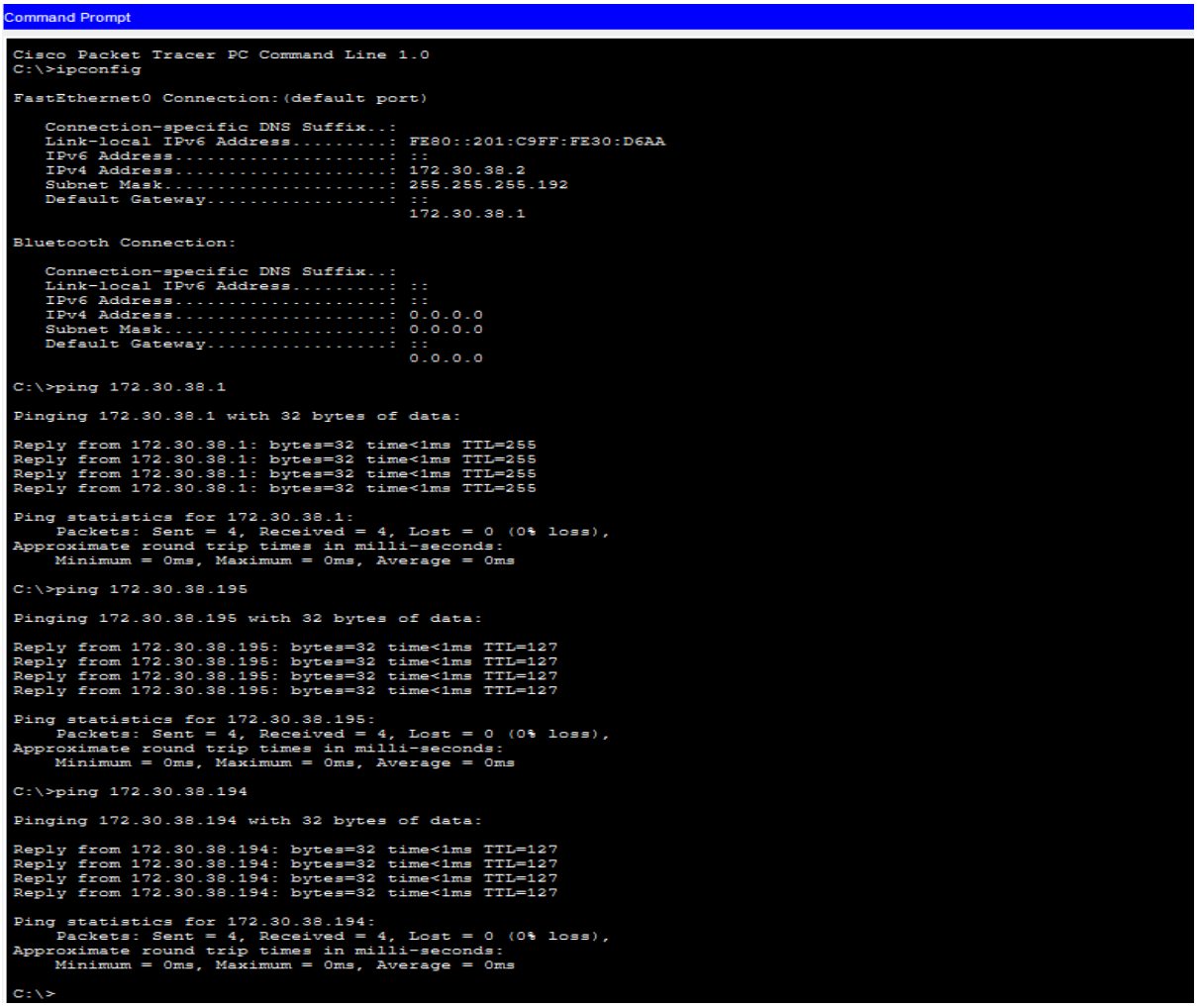
SERVER-01, with IP address 172.30.38.195, provides centralised Dynamic Host Configuration Protocol (DHCP) services.

DHCP automatically assigns network settings to client devices, including IP addresses, subnet masks, default gateways and DNS server addresses.

DHCP relay configuration on the core switch allows requests from the configured client VLANs to reach the central server. This reduces the need to configure every client manually.

The DHCP configuration was tested using client devices to check whether they obtained appropriate network settings.

Figure 4: Booking client connectivity and addressing test



3.2 DNS configuration

The central DNS service is configured on SERVER-01 at 172.30.38.195. The internal DNS domain is phenyotours.local.

The website hostname www.phenyotours.co.za is configured to resolve to the web server at 172.30.38.194.

DNS allows clients to access the website using its hostname rather than entering the server's IP address directly.

The DNS configuration was tested by accessing the configured hostname from a client browser.

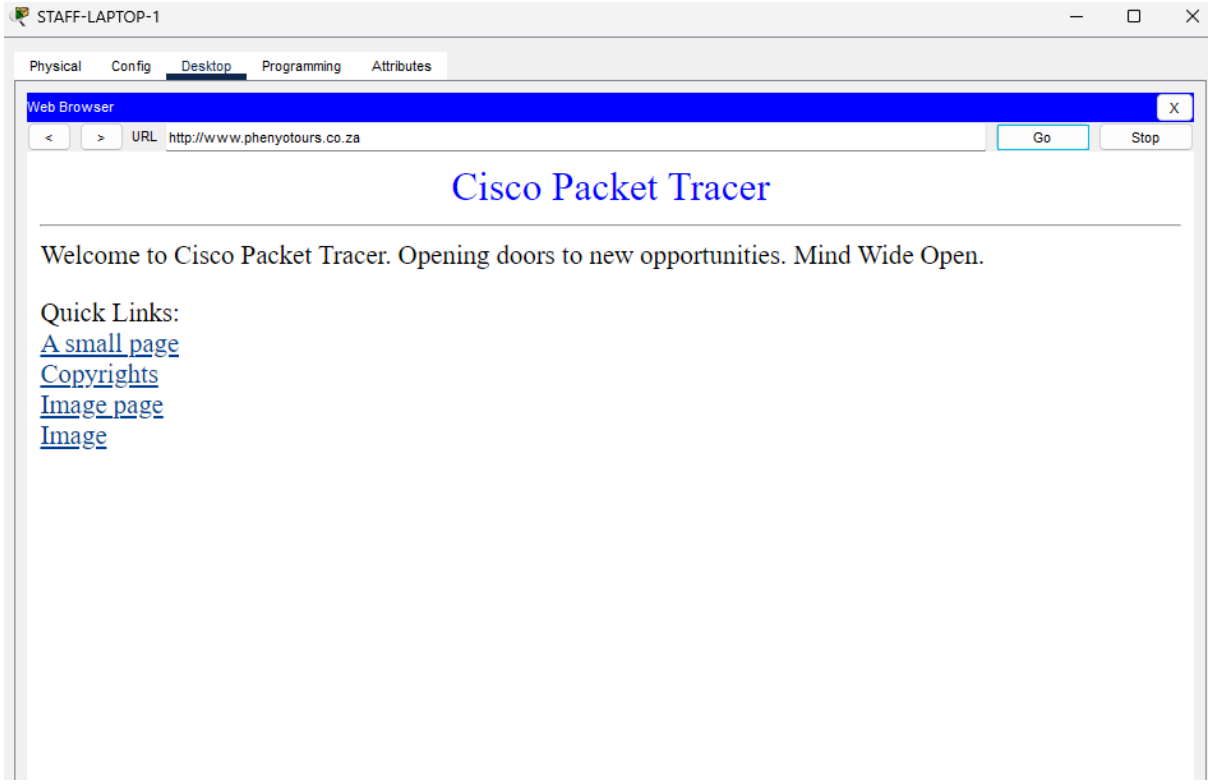
3.3 Web server configuration

SERVER-WEB uses the static IP address 172.30.38.194, with subnet mask 255.255.255.224 and default gateway 172.30.38.193.

HTTP and HTTPS services were enabled on the server. The website is associated with the hostname www.phenyotours.co.za.

During testing, the website was opened through a client browser using the configured hostname. This provides evidence that the relevant DNS and web services were functioning in the tested environment.

Figure 5: Website and DNS verification



4. WIRELESS NETWORKING

The network design includes a wireless LAN controller and wireless access points to provide wireless connectivity.

The planned wireless networks are:

- Phenyo-Staff — staff wireless network.
- Phenyo-Guest — guest wireless network.

Guest wireless traffic is assigned to VLAN 80, with subnet 172.30.39.0/25 and default gateway 172.30.39.1. The wireless configuration includes a management interface for the WLC-3504.

Wireless client testing was used to verify IP address assignment and connectivity to the configured network services.

The presence of a configured WLC management interface does not, by itself, prove that every access point is successfully managed by the controller. Any claim about centralised

wireless management should therefore be supported by additional controller and access-point verification evidence.

5. SECURITY IMPLEMENTATION

5.1 Switch port security

Switch port security was configured on the four departmental access switches: SW-BOOKING, SW-TOUR-OPS, SW-FINANCE and SW-MARKETING.

The following settings were applied to designated access ports:

- Port security enabled.
- Sticky MAC address learning configured.
- Maximum of two MAC addresses per port.
- Violation mode set to restrict.

Sticky MAC learning allows the switch to learn permitted MAC addresses dynamically. The maximum address limit restricts how many MAC addresses are allowed on each protected port.

The restrict violation mode drops traffic from violating MAC addresses and records security violations while keeping the port operational.

Port-security status was checked using Cisco IOS verification commands.

Figure 6: Port-security verification

```
SW-MARKETING#show port-security
Secure Port MaxSecureAddr CurrentAddr SecurityViolation Security Action
             (Count)       (Count)        (Count)
--------------------------------------------------------------------------
        Fa0/1        2          1                0         Restrict
        Fa0/2        2          0                0         Restrict
        Fa0/3        2          1                0         Restrict
        Fa0/4        2          1                0         Restrict
        Fa0/5        2          2                0         Restrict
--------------------------------------------------------------------------
SW-MARKETING#show port-security interface fa0/5
Port Security              : Enabled
Port Status                : Secure-up
Violation Mode             : Restrict
Aging Time                 : 0 mins
Aging Type                 : Absolute
SecureStatic Address Aging : Disabled
Maximum MAC Addresses      : 2
Total MAC Addresses        : 2
Configured MAC Addresses   : 0
Sticky MAC Addresses       : 2
Last Source Address:Vlan   : 0001.9732.42C4:40
Security Violation Count   : 0

SW-MARKETING#
```

5.2 Secure device administration

Security settings were implemented on CORE-L3 and the four departmental access switches.

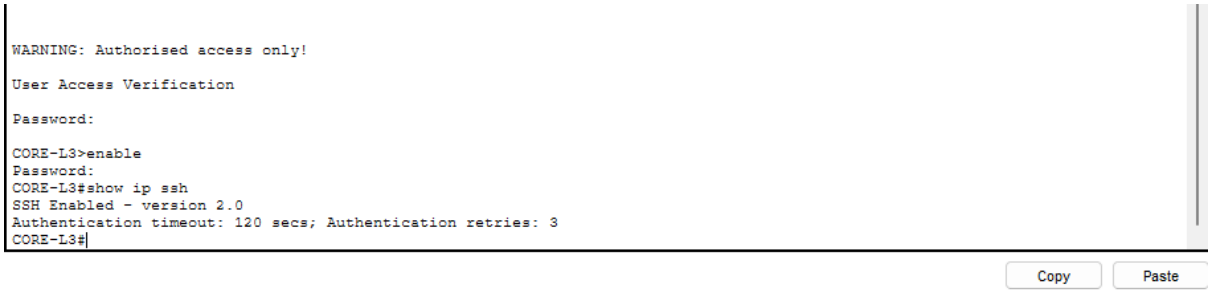
The configured measures include:

- Enable secret for privileged access.
- Password encryption.
- Local administrative account.
- Console authentication.
- Login warning banner.
- RSA key generation.
- SSH version 2.
- VTY access restricted to SSH.

These measures help protect the network devices against unauthorised administrative access and reduce the risk of unapproved configuration changes.

SSH provides encrypted remote administration, unlike unencrypted Telnet. The SSH configuration was verified using the available switch command output.

Figure 7: SSH security verification



6. NETWORK TESTING AND VERIFICATION

Testing was performed to check the operation of the main network components and services.

6.1 VLAN and routing tests

The VLAN configuration and routing table were examined to verify the configured network segments and the availability of routing information.

The core switch provides the configured VLAN gateways and inter-VLAN routing functionality.

6.2 DHCP tests

Client devices were checked to determine whether they received IP addresses and other network settings from the central DHCP server.

The assigned addresses and gateways were compared with the relevant subnet information.

6.3 Connectivity tests

Connectivity tests were performed using client devices and the configured network infrastructure. These tests included gateway and server reachability.

The Booking connectivity screenshot provides supporting evidence for the client connectivity test.

6.4 DNS and web tests

The website hostname was tested from a client browser. Successful access using the hostname provides evidence that the relevant name-resolution and web-server functions were working in the tested environment.

6.5 Security tests

The port-security status was examined to check the configured maximum MAC addresses, sticky learning and violation mode. SSH configuration output was also checked to verify the remote administration security settings.

The available screenshots provide supporting evidence for these tests. They demonstrate the functions shown in the evidence but do not establish that every possible failure condition or security scenario has been tested.

7. CR5: SCALABILITY AND FUTURE GROWTH

The client requires the network to accommodate a 25% increase in users without changing existing network addresses or subnets.

The addressing design uses VLSM to allocate appropriately sized subnets to the departments, servers, guest wireless network and management network. A separate /27 subnet is reserved for future expansion.

The planned usable capacities are:

Network segment	Subnet	Usable host capacity
Booking	/26	62
Tour Operations	/26	62
Finance	/27	30
Marketing	/27	30
Servers	/27	30
Guest Wi-Fi	/25	126
Management	/25	126
Future reserve	/27	30

The addressing plan is intended to provide sufficient capacity for the required growth without renumbering the existing networks.

The /23 address block contains 510 usable host addresses. The design documentation states that 496 usable addresses are allocated, leaving 14 usable addresses unallocated within the block.

8. PROJECT EVIDENCE AND FILE ORGANISATION

The project files are organised into folders for the network configuration, design, documentation, Packet Tracer simulation and testing evidence.

The main deliverables include:

- **PacketTracer:** the simulated network file.
- **Configs:** the network configuration summary.
- **Design:** the network topology diagram.
- **Testing:** screenshots of the topology, VLANs, routing, connectivity, website, port security and SSH.
- **Docs:** the project documentation.

The Packet Tracer file provides the simulation environment in which the network configuration can be inspected and tested.

Project repository:

https://github.com/musanicholas100-arch/CMPG325-2026-063-Phenyo-Cultural-Tours/tree/main

9. CONCLUSION

The Phenyo Cultural Tours network implementation applies key computer networking concepts, including VLAN segmentation, VLSM addressing, inter-VLAN routing, centralised DHCP and DNS services, web hosting, wireless connectivity and switch security.

The configuration summary and available testing screenshots provide evidence of the implemented network features and the verification activities performed.

The use of separate VLANs improves network organisation, while secure administration and port-security settings help protect the network infrastructure. The addressing design also provides capacity for future growth.

Overall, the project demonstrates the practical application of network design, configuration, security and testing using Cisco Packet Tracer.